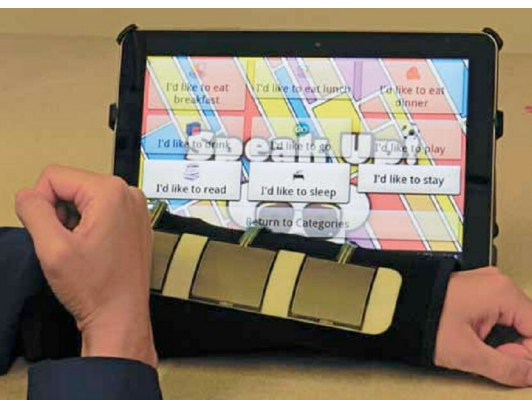


'Special' Technologies Make The World More Accessible To All

Various technologies, products and services are available today that supplement the abilities of specially-challenged people, enabling them to lead life on par with the others. Here we look at some of such technologies from India and outside that serve as an inspiration to those willing to adapt existing technologies or develop new ones for people with special needs



Access4Kids wireless input device

JANANI GOPALAKRISHNAN VIKRAM

According to a recent report in *The Guardian*, titled 'Technology Makes Higher Education Accessible To Disabled Students,' technology plays a huge role in bridging the gap between special and normal people. It cites America's Student E-rent Pilot Project (STEPP) as an example. The programme offers e-textbooks specifically modified for accessibility. In a subsequent survey of 1185 students, 77 per cent reported having saved money by renting their textbooks, and 80 per cent who needed

an accessible textbook were satisfied with the quality of accessibility.

This is just one example of how technology makes life much simpler, more exciting and normal for people with various disabilities. There are a variety of electronic aids like hearing aids, implants, prosthetic limbs, assistive canes and wheelchairs that help special people overcome their disabilities. Today's operating systems are accessible to people with hearing, visual, comprehension and other impairments. Technologies are available that enable challenged people to play great music. Specially-designed user interfaces and devices help autistic

children to use touchscreen-equipped computers. Advanced text-to-speech converters and a vast library of audio books enable visually-challenged people to complete higher education smoothly. Speech, auto-recognition and other technologies enable special people to use ATMs and other facilities in much the same way as others do.

Identifying the need, assessing the requirements in detail, developing suitable technologies and testing them is an intense task. It is quite different from developing normal technologies. Designers and developers of special technologies need to have an empathy and complete understanding of the requirements, behaviour and usage patterns of the people concerned. They need to keep the product cost-effective and simple. The product needs to be tested with the target users, with complete patience and understanding of their sentiments. Whenever possible, the product must also be adaptable to the needs of people with other kinds of impairments. (In fact, this is a consideration that developers of mass-market products should also keep in mind.)

Developing technologies for special people requires as much thoughtfulness and people skills as technical prowess. The results, of course, are worth the efforts, as such projects offer total 100 per cent personal satisfaction.

We salute the special people of this world, who march ahead, succeed and make things happen despite all odds. And, the technologists who help them do so. Whilst it is not possible to speak of all or even most of such technologies, here we have chosen and described a handful of them that seemed different.